

Sparse Reduced Rank Regression: A Simulation Study

Name: Lucy McLaughlin

Supervisor: Dr. Yue Zhao

Department of Mathematics

University of York

April 2026

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1 Simulation Design

1.1 Simulation setup

All simulations in the study were conducted in the statistical programming language, R [1], using the *rrpack* package [2]. The data were generated using the `rrr.sim2()` function [3] which follows the model

$$\mathbf{Y} = \mathbf{X}\mathbf{C} + \mathbf{E}.$$

The rows of $\mathbf{X} \in \mathbb{R}^{n \times p}$ were generated from a multivariate normal distribution with mean vector zero and covariance matrix $\Sigma_{\mathbf{X}}$:

$$\mathbf{x}_i \sim \mathcal{N}_p(\mathbf{0}, \Sigma_{\mathbf{X}}).$$

The i -th row and j -th column of the covariance matrix were defined as

$$\Sigma_{\mathbf{X}_{ij}} = 0.5^{|i-j|},$$

to introduce moderate correlation between predictors, which is commonly observed in real datasets. With the exception of two scenarios, where the predictors were assumed to be uncorrelated, in these cases $\Sigma_{\mathbf{X}} = \mathbb{I}_p$ (where $\mathbb{I}_p$ is the $p \times p$ identity matrix).

The rows of the error matrix, $\mathbf{E} \in \mathbb{R}^{n \times q}$ were generated from

$$\epsilon_i \sim N_q(\mathbf{0}, \sigma^2 \mathbb{I}_q),$$

where $\mathbb{I}_q$ is the $q \times q$ identity matrix. The variance σ^2 was chosen so that the signal-to-noise ratio (SNR) was equal to 1, defined as

$$\text{SNR} = \frac{\text{Var}(\mathbf{X}\mathbf{C})}{\text{Var}(\mathbf{E})},$$

where $\text{Var}(\mathbf{X}\mathbf{C})$ and $\text{Var}(\mathbf{E})$ denote the total element-wise variances of the signal and error matrices respectively.

The errors were assumed to be independent across response variables to simplify the simulation setting and to avoid additional dependence that may affect the evaluation of the coefficient matrix estimation. Although this assumption was made, dependence between the responses remains through the component $\mathbf{X}\mathbf{C}$.

The coefficient matrix, $\mathbf{C} \in \mathbb{R}^{p \times q}$ is modelled as

$$\mathbf{C} = \mathbf{B}\mathbf{A}^T,$$

where the first p_0 rows of $\mathbf{B} \in \mathbb{R}^{p \times r}$ were generated from $\mathcal{N}(0, 1)$, and the remaining $p - p_0$ rows were set to zero to induce sparsity. The rows of $\mathbf{A} \in \mathbb{R}^{q \times r}$ were generated

from $\mathcal{N}(0, 1)$. The true rank, r , of $\mathbf{C}$ was set to three to ensure that the reduced rank structure was meaningful without the model being oversimplistic, such as a case where $r = 1$. The number of observations was set to $n = 100$, providing stability for model estimation; and the number of responses was set to $q = 50$, representing a moderately high-dimensional multivariate setting where low-rank structure benefits the model.

Before fitting the estimators, the dataset was split into a training set and a testing set. A random sample of $0.7n$ observations was used for model estimation, giving the training set a sample size of $n_{train} = 70$, while the remaining 30% of observations were used to evaluate predictive performance, resulting in a testing sample size of $n_{test} = 30$.

When $p < n_{train}$, the OLS estimator was obtained using its closed-form solution. In the scenarios where $p > n_{train}$, the OLS estimator was computed using the Moore-Penrose pseudoinverse, allowing estimation in high-dimensional settings where $\mathbf{X}^T \mathbf{X}$ is singular.

The RRR estimator was obtained using the `rrr.fit()` function from the *rrpack* package. The model was fitted using the training data, and the rank constraint was set to the true rank $r = 3$, as defined in the data-generating process.

The SRRR estimator was obtained using the `srrr()` function from the *rrpack* package. A group-lasso-type penalty was applied to induce row-wise sparsity in the coefficient matrix. The rank was fixed at $r = 3$ in the function, and the tuning parameter controlling the sparsity penalty was selected using the *BIC*.

1.2 Evaluation Criteria

To assess robustness, multiple simulation scenarios were considered: four baseline scenarios varied the parameters p and p_0 , and two additional scenarios with uncorrelated predictors were introduced for comparison in the setting where $p > n_{train}$. Results were averaged over 100 repetitions of the simulations, with a fixed random seed used so that the simulation draws were reproducible.

The first two scenarios consider settings where $p < n_{train}$, allowing comparison between OLS, RRR and SRRR. Scenario 1 set the number of predictors to $p = 65$ and the number of relevant predictors to $p_0 = 30$. This scenario was used to provide a general comparison of all three methods: OLS, RRR and SRRR, where the coefficient matrix was both low-rank and sparse. Scenario 2 set $p = 65$ and $p_0 = 65$, meaning all the predictors in the model were relevant and therefore the true coefficient matrix in this scenario was not sparse. This scenario was used to test the robustness of SRRR, as the sparsity assumption underlying the method is not satisfied.

The remaining two scenarios consider $p > n_{train}$, which violates the invertibility condition required for the classical OLS estimator. Therefore, the OLS was computed

using the Moore-Penrose pseudoinverse for completeness. Scenario 3 set $p = 200$ and $p_0 = 75$, representing a moderately sparse high-dimensional setting. This scenario was used to assess model performance when sparsity is present, but a substantial number of predictors remain relevant, so that SRRR is not unduly favoured. Scenario 4 set $p = 200$ and $p_0 = 10$, producing a case with a highly sparse true coefficient matrix $\mathbf{C}$. This scenario tested how the estimator methods performed when the model has many irrelevant predictors.

To further investigate the role of predictor dependence, two additional scenarios were constructed to mirror scenarios 3 and 4, but with uncorrelated predictors. Thus, Scenario 5 set $p = 200$ and $p_0 = 75$, and Scenario 6 set $p = 200$ and $p_0 = 10$. This separates the effect of correlation from the effect of sparsity and model structure, providing insight into whether predictor correlation affects SRRR's ability to distinguish between relevant and irrelevant predictors. In addition, comparison across correlated and uncorrelated scenarios helps assess whether dependence among predictors influences the overall behaviour of each method.

Method performance was evaluated through prediction accuracy and signal recovery using the testing set, while coefficient estimation accuracy was assessed by comparing the estimated coefficient matrix $\hat{\mathbf{C}}$ with the true coefficient matrix $\mathbf{C}$. Let $\mathbf{X}_{test} \in \mathbb{R}^{n_{test} \times p}$ and $\mathbf{Y}_{test} \in \mathbb{R}^{n_{test} \times q}$ denote the predictor and response matrices for the testing set. The following performance metrics were used:

- **Mean Squared Prediction Error (MSPE):**

$$\frac{1}{n_{test}q} \left\| \mathbf{Y}_{test} - \mathbf{X}_{test} \hat{\mathbf{C}} \right\|_F^2,$$

which measures the average prediction error of the fitted model.

- **Signal Prediction Error:**

$$\frac{1}{n_{test}q} \left\| \mathbf{X}_{test} \mathbf{C} - \mathbf{X}_{test} \hat{\mathbf{C}} \right\|_F^2,$$

which evaluates the model's ability to recover the true underlying signal.

- **Estimation Error:**

$$\left\| \hat{\mathbf{C}} - \mathbf{C} \right\|_F^2,$$

which quantifies the accuracy of the estimated coefficient matrix relative to the true coefficient matrix.

In addition to reporting the average values of these metrics, the median and interquartile range (IQR) were also computed to provide more robust summaries of performance across simulation runs. This is particularly useful in the present study, as the

variability of OLS, RRR, and SRRR may differ substantially across both low- and high-dimensional settings. In particular, the OLS pseudoinverse solution is expected to exhibit unstable performance when $p > n_{train}$, while RRR and SRRR depend on accurate recovery of the underlying rank structure, with SRRR additionally relying on variable selection through penalisation. The median provides a measure of central tendency that is less sensitive to extreme runs, while the interquartile range measures variability and therefore offers insight into estimator stability.

For each scenario, the number of predictors selected by the SRRR estimator was recorded, along with the **true positive rate (TPR)** and **false positive rate (FPR)**, defined as:

- **True Positive Rate (TPR):**

$$\frac{\text{Number of correctly selected relevant predictors}}{\text{Total number of relevant predictors}}$$

- **False Positive Rate (FPR):**

$$\frac{\text{Number of incorrectly selected irrelevant predictors}}{\text{Total number of irrelevant predictors}}$$

These measures provide insight into the variable selection performance of SRRR, allowing evaluation of its ability to correctly identify relevant predictors while avoiding the inclusion of irrelevant ones.

2 Simulation Results

2.1 $p < n_{train}$ Scenarios

The average prediction and estimation error results from the two scenarios, where $p < n_{train}$, are summarised in Table 1. OLS produces significantly larger errors than both RRR and SRRR across both scenarios. SRRR achieves the lowest prediction and estimation errors overall, suggesting that SRRR is generally the most effective method when $p < n_{train}$.

Table 1: Error results for scenarios where $p < n_{train}$ over 100 runs. For each metric, the mean, median, and interquartile range (IQR) are reported.

Scenario	Method	MSPE			Signal Prediction Error			Estimation Error		
		Mean	Median	IQR	Mean	Median	IQR	Mean	Median	IQR
1 ($p = 65, p_0 = 30$)	OLS	271.41	189.57	180.21	254.80	177.10	173.51	25066.36	17950.91	17390.26
	RRR	33.34	23.28	20.35	17.44	11.42	12.11	1773.09	1085.80	1375.20
	SRRR	19.03	14.42	11.10	3.00	2.35	2.06	272.97	227.54	178.30
2 ($p = 65, p_0 = 65$)	OLS	668.67	516.69	536.07	632.50	475.71	494.70	61705.13	45447.66	51034.06
	RRR	79.83	55.25	56.63	43.00	25.37	32.53	4204.83	2660.76	3235.15
	SRRR	67.75	52.49	55.24	30.73	22.10	26.19	2928.21	2023.52	2703.59

In Scenario 1, where $p = 65$ and $p_0 = 30$, the signal prediction error distributions for each method are shown in Figure 1. A large reduction in error is observed when moving from OLS to RRR, highlighting the benefit of imposing a low-rank structure on the coefficient matrix even when $p < n$. SRRR further reduces this error and obtains the lowest variability across simulation runs. This indicates that the additional sparsity constraint is more accurate in recovering the true signal component $\mathbf{XC}$ when the true coefficient matrix is sparse.

This improvement is also evident from the distributional summaries. The median signal prediction error for SRRR is substantially lower than that of RRR and OLS. Furthermore, SRRR exhibits a much smaller interquartile range compared to RRR and OLS, indicating significantly reduced variability. These differences are illustrated in the box plots in Figure 1, where SRRR exhibits both a lower median and a more compact spread.

All box plots are presented on a logarithmic scale to improve visual comparison across methods, particularly where error values are highly skewed or differ substantially in magnitude. Although this compresses larger values and may visually understate the differences in variability between the methods, Figure 1 still clearly shows that the additional sparsity constraint in SRRR leads to more accurate and stable recovery of the true signal component $\mathbf{XC}$ when the coefficient matrix is sparse.

Figure 1: Box plots of signal prediction errors across 100 simulation runs for Scenario 1 ($p < n_{train}, p = 65, p_0 = 30$), shown on a logarithmic scale.

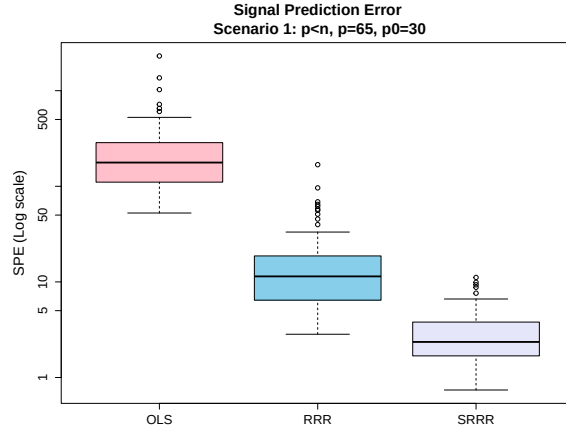
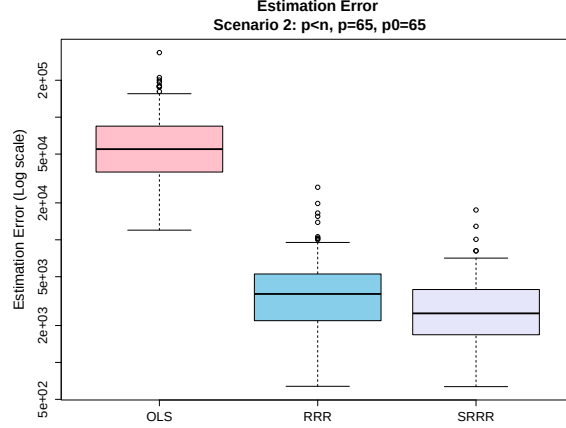


Figure 2: Box plots of estimation errors across 100 simulation runs for Scenario 2 ($p < n$, $p_0 = p = 65$), shown on a logarithmic scale.



In Scenario 2, where $p = p_0 = 65$, the prediction and estimation errors are larger than in Scenario 1. This is expected, as the increase in model complexity leads to a more challenging estimation problem. A notable result in this scenario is that SRRR still outperforms RRR across all error measures, despite the absence of sparsity. One might expect SRRR to perform similarly to RRR when all predictors are relevant. However, while the differences between prediction errors are relatively small, the median estimation error for SRRR is lower than that of RRR, indicating improved estimation accuracy. In addition, the interquartile range of SRRR is smaller, suggesting SRRR remains more stable than RRR even in the absence of sparsity, as illustrated in Figure 2.

Although SRRR achieves a lower estimation error compared to RRR in Figure 2, the gap is noticeably smaller than in Figure 1. This is explained by the fact that all predictors are relevant in Scenario 2, and therefore there are no irrelevant predictors for SRRR to remove. As a result, the slight improvement in estimation ability is likely due to the additional regularisation induced by the group-lasso-type penalty, which can reduce estimation variance even when all predictors remain active. This highlights that SRRR performs best when sparsity is present in the true coefficient matrix.

2.2 $p > n_{train}$ Scenarios

The results in the scenarios where $p > n_{train}$ are given in Table 2. In these settings, the OLS estimator is obtained using the Moore-Penrose pseudoinverse. While the pseudoinverse provides a valid solution, it is known to be unstable in high-dimensional settings and therefore is included primarily as a baseline comparator.

Table 2: Error results for correlated scenarios ($p > n_{train}$) over 100 runs. For each metric, the mean, median, and interquartile range (IQR) are reported.

Scenario	Method	MSPE			Signal Prediction Error			Estimation Error		
		Mean	Median	IQR	Mean	Median	IQR	Mean	Median	IQR
3 ($p = 200, p_0 = 75$)	OLS	151.21	131.69	69.65	101.12	95.24	32.83	9999.42	9461.29	2683.08
	RRR	126.07	112.22	47.40	75.92	72.70	18.99	7536.60	7342.58	1498.37
	SRRR	116.02	103.42	52.83	65.60	63.88	18.94	6365.00	6073.48	1509.87
4 ($p = 200, p_0 = 10$)	OLS	16.17	15.07	8.39	12.25	11.96	4.93	1205.98	1194.84	391.51
	RRR	14.24	13.60	6.30	10.30	9.68	4.17	1012.97	996.33	318.15
	SRRR	4.25	2.95	3.54	0.36	0.24	0.31	31.56	21.04	27.40

Across these scenarios, Table 2 shows the SRRR method remains favourable with lower prediction and estimation errors than both OLS and RRR. The OLS method performs substantially worse with large errors, reflecting its poor performance in high-dimensional settings due to lack of structural constraints. However, the extent to which SRRR outperforms RRR depends strongly on the level of sparsity of the true coefficient matrix.

Scenario 3 represents a moderately sparse setting, with $p = 200$ and $p_0 = 75$. In this case, the limitations of OLS in high-dimensional settings are particularly evident in the estimation error, as illustrated in Figure 3. OLS produces a substantially higher median estimation error compared to RRR. In addition, OLS exhibits a much larger interquartile range, compared to RRR, indicating significantly greater variability. Therefore, OLS is clearly unstable and produces highly variable estimates in high-dimensional settings.

Moreover, in this scenario, SRRR provides a consistent but relatively modest reduction in errors compared to RRR. In the moderately sparse setting of Scenario 3, many predictors remain relevant, so the sparsity penalty provides only limited improvement beyond the low-rank structure already imposed by RRR. For example, the median signal prediction error decreases from approximately 75.50 for RRR to 63.56 for SRRR, while the interquartile range reduces only slightly (from approximately 22.32 to 20.69), indicating some improvement in typical performance and stability, but less pronounced gains.

This contrasts with Scenario 4, which represents a highly sparse setting with $p = 200$ and $p_0 = 10$. In this scenario, SRRR substantially outperforms both RRR and OLS on all error measures. The median signal prediction error reduces sharply from approximately 9.87 for RRR to 0.26 for SRRR, while the interquartile range decreases from approximately 5.11 to 0.21. These differences are clearly illustrated in Figure 4, with a markedly tighter spread for SRRR, indicating more stable performance across simulation runs. The differences between the scenarios illustrate how the advantages of SRRR become increasingly pronounced as sparsity strengthens.

Figure 3: Box plots of estimation error across 100 simulation runs for Scenario 3 ($p > n_{\text{train}}, p = 200, p_0 = 75$), shown on a logarithmic scale.

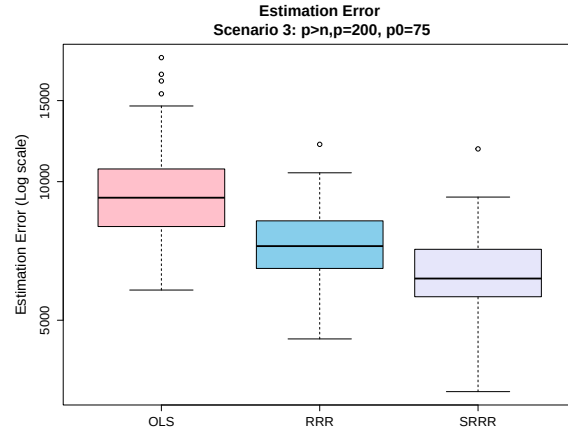
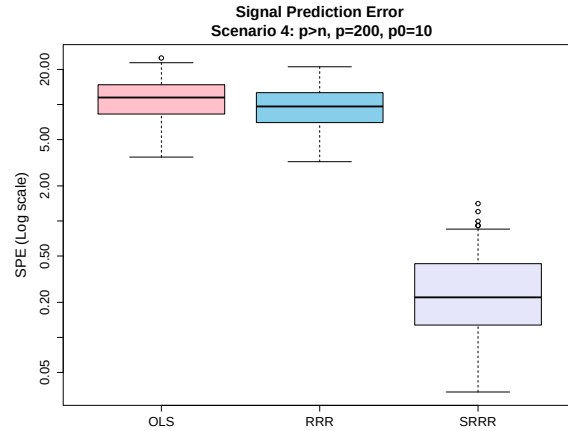


Figure 4: Box plots of signal prediction error across 100 simulation runs for Scenario 4 ($p > n_{\text{train}}, p = 200, p_0 = 10$), shown on a logarithmic scale.



2.3 Additional Uncorrelated Scenarios

The results for the additional scenarios with uncorrelated predictors, Scenario 5 and Scenario 6, are summarised in Table 3. A consistent pattern appears across all methods. In general, the removal of correlation leads to a deterioration in prediction performance. However, the effect on estimation accuracy is less pronounced, with only modest differences observed across scenarios.

Table 3: Error results for uncorrelated scenarios ($p > n_{train}$) over 100 runs. For each metric, the mean, median, and interquartile range (IQR) are reported.

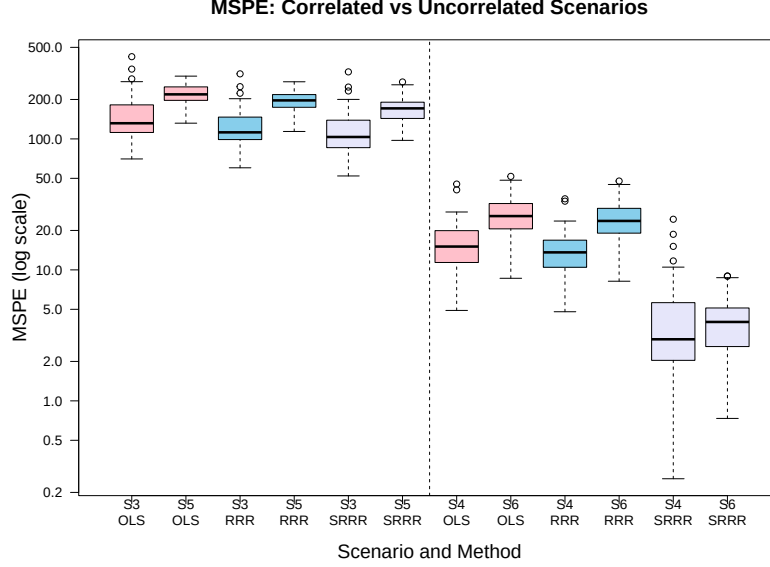
Scenario	Method	MSPE			Signal Prediction Error			Estimation Error		
		Mean	Median	IQR	Mean	Median	IQR	Mean	Median	IQR
5 ($p = 200, p_0 = 75$)	OLS	219.68	218.80	52.33	172.03	172.90	40.20	8609.82	8422.90	1460.34
	RRR	196.02	196.81	43.43	148.16	147.71	38.25	7419.27	7303.52	1342.43
	SRRR	170.90	171.02	46.52	123.05	117.79	41.62	6159.55	6010.36	1551.78
6 ($p = 200, p_0 = 10$)	OLS	26.15	25.75	11.37	22.34	21.46	9.84	1097.72	1096.52	399.78
	RRR	24.27	23.64	10.38	20.46	19.52	9.56	1003.59	998.31	367.02
	SRRR	4.17	4.00	2.45	0.37	0.32	0.25	17.87	16.30	12.00

Comparing the moderately sparse settings in Scenario 3 and Scenario 5, all three methods exhibit higher prediction errors in Scenario 5 with uncorrelated predictors. For example, the median signal prediction error for SRRR increases substantially, indicating that predictor correlation may aid predictive performance. This is also evident in Figure 5, where the median mean squared prediction error is higher in Scenario 5 than in Scenario 3. In contrast, the estimation error shows only minor differences, with slightly lower mean values but a similar median and interquartile range. This suggests that predictor dependence has a limited effect on coefficient estimation.

Similarly, the distributional summaries show that variability behaves differently across metrics. While the interquartile range of signal prediction error increases in the uncorrelated case, the interquartile range for mean squared prediction error is generally smaller, and estimation error variability remains broadly similar across the two scenarios. This indicates that predictor correlation influences predictive stability more than coefficient estimation stability.

A similar pattern is observed in the highly sparse scenarios where $p_0 = 10$. The uncorrelated scenario again produces higher prediction errors across OLS and RRR, while SRRR maintains a strong performance. This is evident in the comparison of the median mean squared prediction error values in Figure 5, where SRRR remains substantially lower overall. The differences in estimation errors remain relatively small across the scenarios, and variability does not show a consistent increase in the uncorrelated setting. This suggests that, in highly sparse settings, the benefits of explicit variable selection may be more influential than predictor correlation in determining model performance.

Figure 5: Mean Squared Prediction Error (MSPE) across 100 simulation runs in scenarios where $p > n_{train}$. Box plots compare the performance of OLS, RRR, and SRRR between the uncorrelated and correlated scenarios.



2.4 SRRR Predictor Selection Performance

The recovery of sparsity by SRRR across the four scenarios is summarised in Table 4. By comparing the number of predictors removed by SRRR with the true number of irrelevant predictors, it can be seen that SRRR eliminates a substantial proportion of irrelevant predictors.

Table 4: Average SRRR results across all simulation scenarios over 100 runs. The table reports number of true irrelevant predictors, number of predictors removed, True Positive Rate (TPR), and False Positive Rate (FPR).

Scenario	p	p_0	True Irrelevant Predictors	Predictors removed	TPR	FPR
1	65	30	35	20.74	0.99	0.42
2	65	65	0	0.48	0.99	NA
3	200	75	125	82.28	0.80	0.46
4	200	10	190	137.72	1.00	0.28
5	200	75	125	79.63	0.81	0.48
6	200	10	190	122.12	1.00	0.36

In the correlated scenarios, the ability of SRRR to remove irrelevant predictors improves as the level of sparsity increases. This is particularly evident in Scenario 4, where the true coefficient matrix is highly sparse. In this scenario, SRRR removes

many irrelevant predictors while the true positive rate shows accurate recovery of relevant predictors. In contrast, in Scenario 2, when all predictors are relevant SRRR removes almost no predictors on average, demonstrating that the method does not enforce sparsity when none is present.

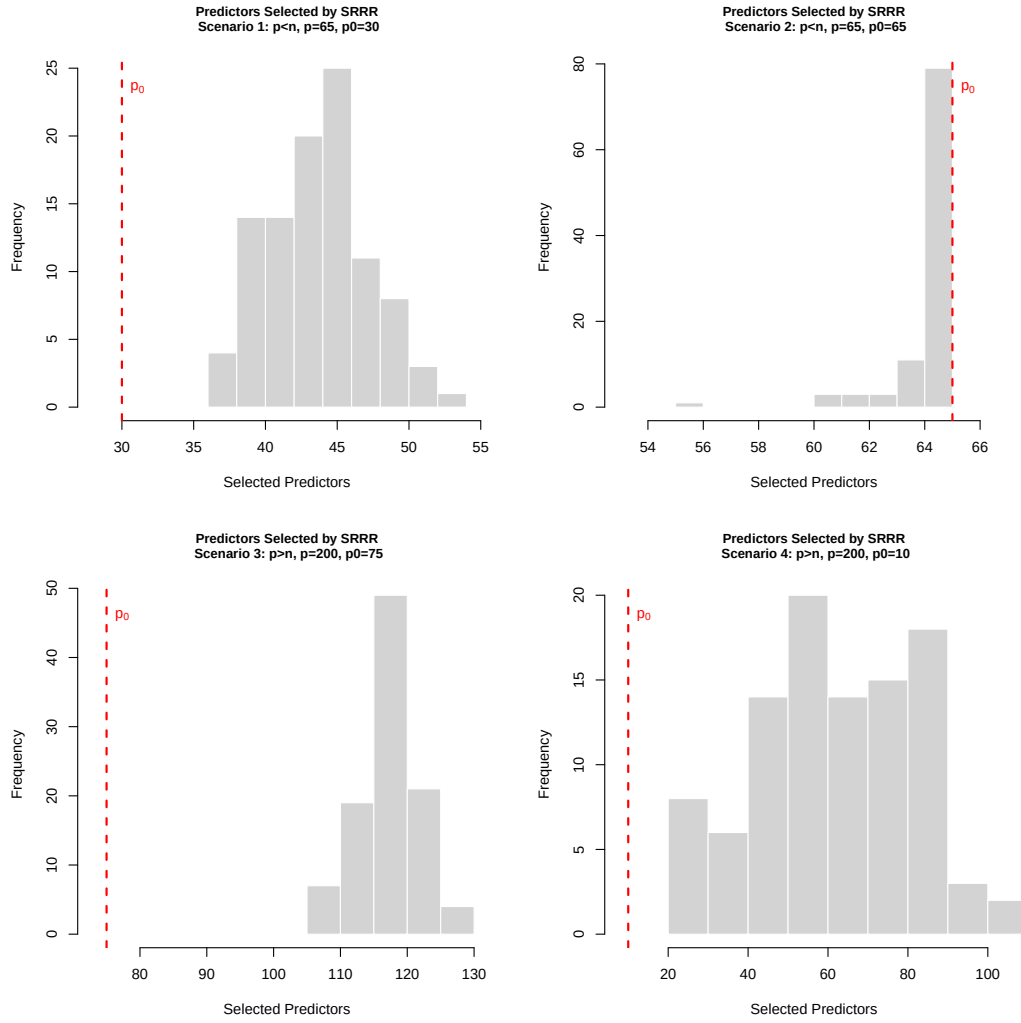
In all cases the true positive rate is very high, indicating that the relevant predictors are almost always correctly identified. However, a moderate false positive rate is observed in some scenarios. This may be partly due to the use of BIC for selecting the sparsity tuning parameter, which can favour slightly larger models. While one may assume that correlation among predictors can contribute to the difficulty of identifying relevant predictors compared to irrelevant ones, the results from the uncorrelated scenarios suggest that this is not the primary driver of the false positives.

When comparing the variable selection performance of SRRR in the uncorrelated scenarios (scenarios 5 and 6) to their corresponding correlated scenarios (scenarios 3 and 4), it is observed that the false positive rate does not decrease when correlation is removed. In the moderately sparse setting the false positive rate remains relatively similar, while in the highly sparse setting it increases (from approximately 0.28 to 0.36). This suggests that the inclusion of irrelevant predictors is more strongly influenced by the regularisation of the model and the method choice for selecting the tuning parameter, than by predictor correlation alone.

The distribution of the number of predictors selected by SRRR in the baseline scenarios are illustrated in Figure 6. The spread of the histograms indicates that the number of predictors selected by SRRR varies across the simulation runs. This is reflective of the stochastic nature of the simulated datasets, where different random realisations of the data lead to small differences in the estimated coefficients, and consequently the predictors identified as relevant by SRRR.

In the correlated scenarios where sparsity is present (scenarios 1, 3 and 4), SRRR consistently selects more predictors than the true number of relevant predictors. However, the difference is not excessively large, and the high true positive rates demonstrate that the models do correctly identify the majority of true predictors.

Figure 6: Histograms of the number of predictors selected by SRRR across simulation runs.



2.5 Overall Results

Overall, the simulations demonstrate that SRRR performs strongly relative to OLS and RRR when the true coefficient matrix is low-rank and sparse. Across the scenarios considered, SRRR consistently achieves the lowest or near-lowest prediction and estimation errors, while also identifying the majority of relevant predictors. The benefits of both SRRR and RRR over OLS are clear in all scenarios, while the benefits of SRRR over RRR become more evident as the level of sparsity in the true coefficient matrix increases.

References

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